

SECURE AI HACKATHON

By Seattle Data AI & Security Community

Participant Guide

HACKATHON DURATION: 48 HOURS

Build a practical AI-powered solution to a real-world problem using data, evidence, safety, and human control.

Beginner-friendly entry points

Optional advanced layers

Working prototype first

Core principle

This is a problem-solving hackathon, not an architecture contest. A simple solution that works, uses data correctly, and demonstrates a clear trust or safety mechanism is stronger than a complex design with no functional product.

Key Terms and Conditions

- Substantial new functionality must be developed during the 48-hour event; pre-existing work must be disclosed.
- Open-source tools, templates, datasets, and third-party or proprietary APIs may be used as disclosed dependencies and do not count as hackathon work.
- Teams must comply with applicable licenses and may use only public, licensed, organizer-provided, participant-owned, or synthetic data.
- Participants retain ownership of their original work. Organizers may review, judge, display, record, and promote submissions for event-related purposes.
- Stolen code, unauthorized data, exposed credentials, malicious functionality, or misleading submissions may be disqualified. Organizer decisions are final.

For students and early-career professionals

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Hackathon Overview

Purpose, participant expectations, and how the event works

Purpose

The Data, AI & Security Hackathon challenges teams to create a working prototype that solves a clear user problem. Projects should use data meaningfully, apply AI to provide useful assistance, and include at least one mechanism that improves trust, safety, privacy, evidence, or human control.

Teams are encouraged to start with a focused use case and a simple end-to-end workflow. Advanced architecture and additional features may be added after the core solution works reliably.

What participants should achieve

Solve a real problem Design for a specific user such as a student, analyst, recruiter, organizer, customer, security analyst, or team member.	Use data meaningfully Work with a CSV, logs, documents, notes, form submissions, email samples, or a synthetic dataset.
Apply AI with purpose Use AI for classification, summarization, question answering, extraction, recommendations, report generation, or action drafting.	Build trust into the workflow Include evidence, approval, privacy masking, confidence labels, safe refusal, logging, or another visible control.

How the hackathon works

1. Choose one of the three tracks and identify the user you want to help.
2. Define one specific problem and one realistic input or dataset.
3. Build the minimum end-to-end flow before adding advanced features.
4. Add at least one trust, safety, privacy, evidence, or human-review mechanism.
5. Test the solution with realistic and failure-case examples.
6. Prepare a concise demonstration of the problem, workflow, architecture, safety layer, and result.

Guide contents

Section	What it covers
1	Hackathon overview and participation model
2	General project requirements
3	Track overview and track-selection guidance
4	Track 1 - Security Reasoning Agent
5	Track 2 - Trustworthy Data Analysis
6	Track 3 - Human-in-the-Loop AI Assistant
7	Judging criteria and architecture expectations
8	Build guidance and project-readiness checklist
9	Official 48-hour agenda, checkpoints, deadlines, and judging schedule
10	Submission form, public repository, ZIP package, video, data/API requirements, and final checklist

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General Project Requirements

The minimum standard that applies to every track

What counts as a valid project?

Minimum requirement	Participant expectation
A real user problem	The project must help a clearly identified user or group with a specific problem.
A supported data input	Use a CSV, log file, text document, notes, email sample, form submission, or synthetic dataset.
Meaningful AI assistance	The AI component must perform a useful task such as summarization, classification, Q&A, recommendations, extraction, report generation, or action drafting.
A visible trust or control feature	Include at least one mechanism such as evidence display, human approval, PII masking, confidence labels, safe refusal, or an audit log.
A working demonstration	A web app, notebook, dashboard, Streamlit or Gradio app, chatbot, simple React app, or clickable prototype is acceptable.

Prompt-only submissions are not sufficient

Required application layer

A prompt may be one component of the project, but it cannot be the entire submission. The solution must include at least one additional layer such as calculations, data validation, evidence display, human approval, privacy masking, logging, guardrails, or test cases.

Core and optional layers

Layer type	Expectation	Example
Core layers	Minimum components required for a useful end-to-end prototype.	Input -> AI assistance -> evidence or review -> final result
Optional advanced layers	Enhancements that improve technical depth, safety, reliability, or usability.	Guardrails, evaluation, audit logs, access control, dashboards, deployment

General final-demo expectations

- Show the project working from input to output using real or realistic sample data.
- Explain what the AI does and what parts are calculated, verified, reviewed, or controlled by the application.
- Demonstrate the required trust, safety, privacy, evidence, or human-control mechanism.
- Explain the architecture clearly without making architecture the main product.
- Show at least one realistic success case and one limitation, uncertainty, or failure-case behavior.

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Choose Your Track

Select the track that best matches your team, problem, and experience level

Track	Recommended level	Primary focus	Core outcome
1. Security Reasoning Agent	Intermediate to advanced	Security + AI	Analyze security signals and produce evidence-backed incident guidance.
2. Trustworthy Data Analysis	Beginner friendly	Data + AI	Answer questions from data using verified calculations, evidence, and basic safety checks.
3. Human-in-the-Loop AI Assistant	Beginner friendly	AI + Control	Draft or recommend actions while requiring human review before finalization.

Track-selection guidance

Choose Track 1 when...

Your team wants to work with phishing messages, logs, alerts, cloud warnings, or security incidents and can explain evidence behind a risk assessment.

Choose Track 2 when...

Your team wants an approachable data project that calculates answers from a dataset and shows the chart, table, or calculation behind every result.

Choose Track 3 when...

Your team wants to build a practical AI workflow that drafts or recommends actions but keeps a human in control before anything is finalized.

Start simple in every track

A focused prototype that works end to end is the priority. Add advanced layers only after the minimum workflow is stable.

What all three tracks have in common

- A clearly defined user and problem.
- A realistic data or text input.
- A useful AI-generated output.
- A visible evidence, trust, safety, privacy, or approval layer.
- A working demonstration that participants can explain clearly.

Do not over-engineer early

Build the smallest useful path first. Once the input-to-output flow works, improve reliability, safety, user experience, and technical depth.

1 Security Reasoning Agent

Intermediate to Advanced | Security + AI | Best for experienced teams

Challenge statement

Build an AI assistant that analyzes suspicious activity, logs, alerts, phishing messages, or security incidents and turns them into clear, evidence-backed guidance.

A beginner entry point may classify one suspicious email or a simple log sample and explain the red flags. An advanced solution may correlate alerts, build an incident timeline, score severity, and recommend response steps.

Solution expectations

The solution must do more than summarize a security input. It should connect its conclusion to visible evidence, avoid unsupported claims, and communicate uncertainty when the available data is insufficient.

Required workflow

Core layer	What teams build	Example
1. Input	Accept one type of security input.	Suspicious email, login log, alert CSV, or cloud warning.
2. AI analysis	Classify risk and summarize what happened.	Low, medium, or high risk with a short explanation.
3. Evidence	Show the lines, words, or events that support the conclusion.	Sender-domain mismatch or five failed logins in three minutes.
4. Recommended action	Provide next steps without presenting uncertain conclusions as facts.	Block sender, reset password, investigate IP, or escalate to an administrator.

Minimum deliverable

- The user can paste or upload one security sample.
- The system returns a risk level, short summary, supporting evidence, and recommended actions.
- The system includes at least one safety layer: evidence requirement, PII masking, or confidence and uncertainty behavior.

Track 1 continued

Optional enhancements, project ideas, and participant fit

Optional advanced layers

Severity scoring Use a repeatable scoring approach instead of an unexplained AI judgment.	PII masking Hide names, email addresses, IP addresses, secrets, or tokens when displaying results.
Incident timeline Organize related events chronologically so the incident is easier to understand.	False-positive handling Allow the system to return “not enough evidence” rather than forcing a conclusion.
Audit log Record the input, AI output, confidence, and reviewer decision.	

Project ideas and expected demo moments

Project idea	Problem it solves	Strong demo moment
Phishing triage assistant	Helps users decide whether an email appears risky.	Highlights red flags and recommends safe next steps.
Suspicious login analyzer	Identifies unusual behavior in a login log.	Shows a timeline and risk score.
SOC alert summarizer	Turns noisy alerts into a short incident brief.	Groups related alerts and explains priority.
Cloud misconfiguration explainer	Explains risky cloud settings in plain language.	Shows the risky setting and a recommended fix.
Best-fit participants Intermediate and advanced students, security-curious developers, data students comfortable working with logs, and early-career professionals interested in cyber defense.		

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Trustworthy Data Analysis

Beginner Friendly | **Data + AI** | **Accessible to beginners**

Challenge statement

Build an AI assistant or dashboard that allows users to ask questions about a dataset and receive answers that are calculated or verified from the data rather than invented by the model.

Teams may use a clean CSV, a notebook, a Streamlit or Gradio application, or a basic dashboard. The goal is not complex AI architecture; the goal is useful and trustworthy answers from data.

Solution expectations

The application must calculate, query, or verify results using the actual dataset. When the data cannot answer a question, the system should say so instead of generating an unsupported response.

Required workflow

Core layer	What teams build	Example
1. Dataset	Use a clean CSV or sample dataset.	Event registrations, expenses, survey responses, or job listings.
2. User question	Allow the user to ask simple questions or select predefined questions.	Which city has the most attendees?
3. Verified answer	Calculate the result using code or a query, then explain it with AI.	Count rows, group by city, and show the top result.
4. Evidence	Show the chart, table, or calculation behind the answer.	Bar chart, mini table, and plain-language explanation.

Minimum deliverable

- Use one dataset with at least 30 rows or a provided sample dataset.
- Answer at least three user questions using the dataset.
- Show the calculation, chart, or source table behind each answer.
- Include one trust feature: sensitive-field masking, source-table display, confidence label, or unsupported-question refusal.

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Track 2 continued

Optional enhancements, project ideas, and participant fit

Optional advanced layers

Sensitive-field masking Hide names, phone numbers, email addresses, salary, or other private fields before display.	Unsupported-question handling State clearly when the available dataset cannot answer a question.
Natural language to SQL/Pandas Convert user questions into visible query or code steps.	Confidence labels Label answers based on data completeness, quality, or ambiguity.
Report export Generate a summary report containing insights, charts, and supporting data.	

Project ideas and expected demo moments

Project idea	Problem it solves	Strong demo moment
Event insights assistant	Helps organizers understand registration and attendee data.	Answers questions and shows charts from an event CSV.
Student budget analyzer	Helps students understand spending patterns.	Detects the top expense category and suggests savings.
Job market tracker	Helps early-career users analyze job postings.	Shows top skills, locations, and companies.
Survey feedback analyzer	Summarizes ratings and open-ended feedback.	Shows sentiment, common topics, and supporting examples.
Study progress dashboard	Helps students track learning progress.	Shows weak topics and recommends what to revise next.

Best-fit participants

Beginners, first-time hackathon participants, data and business analytics students, and early-career professionals looking for a realistic but approachable AI and data project.

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Human-in-the-Loop AI Assistant

Beginner Friendly | AI + Control | No live APIs required

Challenge statement

Build an AI assistant that drafts, recommends, organizes, or prepares an action while keeping a human in control before anything important is finalized.

Teams do not need live Gmail, Slack, calendar, payment, or publishing integrations. Mock actions and an approval queue are enough to demonstrate the workflow safely.

Solution expectations

A prompt that says “ask before taking action” is not a control system. The application itself must prevent unapproved actions and show a clear approve, edit, or reject step together with a record of the final decision.

Required workflow

Core layer	What teams build	Example
1. User request	Accept a task the user wants help with.	Draft a reply to this email or create tasks from these notes.
2. AI draft	Prepare the output without finalizing the action.	Draft email, task list, schedule suggestion, or post caption.
3. Human review	Allow the user to approve, edit, or reject the draft.	Approve, edit, and reject controls.
4. Action log	Record what the system generated and what the user decided.	Draft approved at 3:10 PM or rejected as too risky.

Minimum deliverable

- The user enters a task or uploads notes or text.
- The AI generates a draft or recommendation.
- The user can approve, edit, or reject the output before finalization.
- The system keeps a simple action log.
- The system includes one safety check such as a risk label, private-information warning, or blocked action.

Track 3 continued

Optional enhancements, project ideas, and participant fit

Optional advanced layers

Risk levels Label proposed actions as low, medium, or high risk before review.	Approval rules Require approval for selected actions and block prohibited actions completely.
Guardrail checks Detect private information, harmful content, or unsupported claims before approval.	Undo or rollback Demonstrate how an approved action could be cancelled or reverted.
Role-based approval Assign different approval permissions to different users.	

Project ideas and expected demo moments

Project idea	Problem it solves	Strong demo moment
Email reply assistant	Drafts replies but requires review before sending.	Shows the approve, edit, reject flow and action log.
Meeting notes to tasks	Converts meeting notes into action items.	The user approves only the correct tasks before saving.
Job application helper	Drafts cover letters or outreach messages.	Blocks submission until the user approves the draft.
Event volunteer coordinator	Suggests volunteer task assignments.	An organizer reviews assignments before publishing.
Social post assistant	Drafts professional social content.	Flags risky or unsupported claims before approval.

Best-fit participants

Beginners, product-minded students, non-security participants, early-career professionals, and teams that want to build a practical AI workflow without complex backend integrations.

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Judging and Architecture Expectations

What participants should prioritize when building and presenting

Participant-facing judging criteria

The criteria and weights below are shared so teams understand the priorities of the hackathon. Detailed scoring ranges, judge instructions, tie-breakers, and internal review guidance are maintained separately for the judging panel.

Judging area	Weight	Participant-facing description
Problem and usefulness	25%	Is the problem clear, relevant, and useful to a real user?
Working prototype	25%	Does the project work end to end with real or realistic sample data?
Data and AI quality	15%	Does the solution use the data correctly and apply AI effectively?
Trust and safety	15%	Does the project include a meaningful guardrail, review step, evidence display, privacy check, or safe refusal?
Architecture clarity	10%	Can the team explain the flow clearly from input through processing, AI, safety, and output?
Demo and storytelling	10%	Can the team communicate the problem, solution, and value through a concise demonstration?

Architecture supports the solution

Architecture is not the assignment

Architecture will be evaluated as evidence that the solution is understandable, reliable, safe, and useful. A clear working flow is more important than a complex diagram.

Architecture questions teams should be ready to answer

Question	Beginner answer is sufficient	Advanced answer example
What is the input?	CSV, text, notes, logs, or an email sample.	Multiple sources, validation, and a normalized schema.
What does the AI do?	Summarizes, classifies, extracts, explains, or recommends.	Uses retrieval, tools, structured outputs, or multi-step reasoning.
How do you know the result is correct?	Show a table, chart, evidence, calculation, or user review.	Use automated tests, benchmark examples, or confidence scoring.
What is the safety layer?	Mask PII, require approval, show evidence, or refuse unsupported answers.	Use guardrails, access control, audit logs, or red-team testing.

A strong demonstration should show

- The user problem and why it matters.
- The input or dataset used by the application.
- The working AI-assisted flow.
- The evidence, calculation, review, privacy, or safety mechanism.
- The final result and its practical value.
- Any important limitation or uncertainty the team discovered.

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Build Guidance and Project Readiness

Recommended technology choices, build phases, and readiness checks

Suggested starter stack

Area	Options
Interface	Streamlit, Gradio, a simple React app, Jupyter Notebook, or a basic dashboard.
Data	CSV, JSON, text files, synthetic logs, sample emails, notes, or provided datasets.
AI	OpenAI, Gemini, Claude, local models, or another available LLM API.
Storage	Optional for beginner demos. CSV files, SQLite, Firebase, or in-memory state are sufficient.
Trust and guardrails	Regex checks, PII masking, confidence labels, approval controls, source tables, evidence display, or manual review.

Recommended build phases

Build phase	What the team should accomplish
Phase 1 - Define	Choose the track, user, problem, sample input, and minimum demonstration path.
Phase 2 - Build the core	Create the basic input-to-AI-output workflow without over-engineering.
Phase 3 - Add trust	Add evidence, calculations, charts, approval, privacy masking, safe refusal, or another required control.
Phase 4 - Test and improve	Test realistic examples, failure cases, and the user experience. Add one optional enhancement if time permits.
Phase 5 - Prepare the demo	Explain the problem, solution, architecture, safety layer, and result in a concise story.

Project readiness checklist

☐ We selected one track and defined a specific user problem.	☐ Our solution uses a real or realistic input or dataset.
☐ The core workflow works from input to final result.	☐ The AI component performs a meaningful task.
☐ Our project includes the minimum deliverables for the selected track.	☐ We implemented at least one visible trust, safety, privacy, evidence, or approval feature.
☐ Our submission is more than a standalone prompt.	☐ We tested a normal case and at least one limitation or failure case.
☐ We can explain our architecture in a simple input-to-output flow.	☐ Our demo clearly shows the problem, solution, safety layer, and result.

Final message to participants

Start small. Build something real.

Make the AI useful, then add one layer that makes the solution safer, more trustworthy, or more controllable. That is the point of this hackathon.

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Hackathon Agenda

Official 48-hour schedule, checkpoints, deadlines, and judging

Official 48-hour schedule

The hackathon runs for 48 hours from Thursday, July 30, 2026 at 10:00 AM to Saturday, August 1, 2026 at 10:00 AM.

Date & Day	Time	Session	Activity
Thursday, July 30	10:00 AM	Introduction and Kickoff	Provide all required resources, rules, tracks, and submission guidance.
Thursday, July 30	6:00 PM	Checkpoint 1	Idea validation
Friday, July 31	10:00 AM	Checkpoint 2	Architecture review
Friday, July 31	3:00 PM	Checkpoint 3	Submission readiness
Friday, July 31	6:00 PM	Soft Deadline	Recommended early project submission
Saturday, August 1	10:00 AM	Hard Deadline	Final project submissions close
Saturday, August 1	11:00 AM	Judging	Parallel judging panels

Teams may submit by the soft deadline and continue updating the project until the hard deadline. The final ZIP file and repository must reflect the final version submitted for judging.



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Submission Requirements

Official form, public repository, ZIP package, video, and data/API rules

Official submission form

All teams must submit through the official form: <https://forms.gle/Yc1HKB8MeKRhVJqM7>

Submission form must include

- Team number, project name, selected track, and team-member details.
- Public GitHub repository link.
- YouTube presentation video link.
- Final project ZIP file.

Public GitHub repository requirements

- The repository must be public and contain the complete final source code and commit history.
- Include a clear README with project overview, setup, installation, execution, and demo instructions.
- Include the presentation document, architecture or workflow diagram, dependencies, dataset sources, and API details.
- Disclose pre-existing components and third-party or proprietary services.
- Do not commit API keys, passwords, tokens, private credentials, or confidential data.

Final project ZIP file

- The ZIP file must be the final exported copy of the same project available in the public GitHub repository.
- It must contain the complete code, README, presentation document, architecture or workflow diagram, dataset/API details, and required configuration instructions.
- The ZIP file and repository must match the final project version at the hard deadline.



Submission Requirements continued

Presentation video

- Upload the presentation video to YouTube and submit a working link.
- The video must be no longer than 10 minutes and must include the project explanation and working demo.
- Cover the problem, target user, solution, architecture or data flow, trust/safety feature, practical result, and key limitations.

Dataset and API requirements

- Use only public, properly licensed, organizer-provided, participant-owned, or synthetic datasets.
- Disclose dataset sources, licenses, APIs, and third-party services used by the project.
- Confidential, private, unlawfully obtained, or unauthorized personal data is prohibited.
- Third-party and proprietary APIs are allowed as disclosed external dependencies. Teams are responsible for provider terms, availability, usage limits, and costs.

Final submission checklist

☐ Official submission form is completed.	☐ Team number, project name, selected track, and team-member details are included.
☐ Public GitHub repository link works without an access request.	☐ YouTube presentation video link works and the video is 10 minutes or less.
☐ Final project ZIP file is uploaded and matches the GitHub repository.	☐ README includes setup, installation, execution, and demo instructions.
☐ Presentation document and architecture or workflow diagram are included.	☐ Dataset, API, third-party service, and pre-existing work disclosures are included.
☐ No credentials, secrets, or confidential data are included.	☐ Submission is completed before the hard deadline.

Submit Your Project

<https://forms.gle/Yc1HKB8MeKRhVJqM7>

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